MILITARY SPECIFICATION

FOR

# RAILWAY CAR, FLAT, DEPRESSED CENTER 56-1/2-INCH GAGE, HEAVY DUTY, 140 TON, 12-WHEEL, DOMESTIC SERVICE

## 1. SCOPE

1.1 Scope. This specification covers heavy duty depressed center flat-cars for domestic service.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-P-3320 | - Painting: Freight and Maintenance Cars. |
| MIL-P-3558 | - Plates, Identification: Locomotives, Railway Cars, and Work Equipment. |

(Copies of specifications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Code of Rules Governing the Loading of Commodities on Open Top Cars.



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Manual of Standards and Recommended Practices:

Mechanical Division:

| | |
|---|---|
| Section A | - Specifications for Materials. |
| M-101 | - Axles, Carbon Steel, Nonheat-Treated. |
| M-107 | - Wheels, Wrought Carbon Steel. |
| M-116 | - Steel, Structural, Shapes, Plates and Bars. |
| M-126 | - Forgings, Carbon Steel. |
| M-201 | - Steel Castings. |
| M-404 | - Unions and Pipe Fittings, 300 Pound Pressure. |
| M-917 | - Journal Roller Bearing Grease. |
| Section D | - Trucks and Truck Details. |
| Section E | - Brakes, and Brake Equipment. |
| Section G | - Wheels. |

Wheel and Axle Manual

Specification No. 2518 - Installation, Freight Car Brake Equipment.

Pamphlet No. 5039-4, Sup. 1 - Single Car Testing Device - Code of Test.

Supplement to the Manual of Standards and Recommended Practices.

Plate B Equipment Diagram
Plate C Equipment Diagram

United States Safety Appliances for All Classes of Cars and Locomotives.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, Illinois 60605.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

| | |
|---|---|
| A36 | - Structural Steel. |
| A53 | - Welded and Seamless Steel Pipe. |
| A113 | - Structural Steel for Locomotives and Cars. |



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| A325 | - High Strength Carbon Steel Bolts for Structural Joints, Including Suitable Nuts and Plain Hardened Washers. |
| A440 | - High Strength Structural Steel |
| A441 | - High Strength Low Alloy Structural Manganese Vanadium Steel. |
| A502 | - Steel Structural Rivets. |

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

## 3. REQUIREMENTS

3.1 Description. The car shall be the depressed center type, 140-ton capacity, 56-1/2-inch gage in accordance with the applicable Association of American Railroads (AAR) requirements. The underframe shall be a one-piece steel casting, mounted on two 6-wheel trucks.

3.1.1 Drawings. The supplier is responsible for preparing his own shop drawings. No deviation from the prescribed dimensions is permissible without prior approval of the contracting officer.

3.2 First article.

3.2.1 First-produced car. The supplier shall furnish for examination and test within the time frame specified in the contract or purchase order one or more cars to prove that his design and production methods will produce cars that comply with the requirements of this specification (see 6.3). Examination and tests shall include those specified herein and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Materials. Materials shall be as specified herein. Materials not specified shall conform to AAR, ASTM, or standard railroad requirements.

3.3.1 Steel. Structural steel shall conform to ASTM A36, A113, A440, A441, or AAR M-116. Castings shall conform to AAR M-201, grade A. Forgings shall be carbon steel and conform to AAR M-126.

3.3.2 Pipe and pipe fittings. Pipe shall be black steel conforming to ASTM A53. Pipe fittings shall conform to AAR M-404.

3.3.3 Bolts, nuts, and rivets. Bolts and nuts shall conform to ASTM A325. Rivet material shall conform to ASTM A502.



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3.3.4 Threaded parts. Threaded parts shall conform to Handbook H28. The American National Fine Thread Series shall be used for threaded parts less than 1/4 inch in diameter. The Unified or American National Coarse Thread Series shall be used for threaded parts 1/4 inch in diameter and larger; however, the American National Fine Thread Series may be employed for these sizes where applicable. All airbrake pipe threads shall conform to American National Taper Pipe Thread Series.

3.4 Substitution of materials. Materials not conforming to the specifications referenced herein shall not be substituted; (1) unless such substitutions are approved; (2) when it is clearly demonstrated that an improvement in operating characteristics, a savings in weight, conservation of critical or strategic materials, or a reduction in cost without sacrifice or reliability can be accomplished thereby; and (3) that such substitutions will not preclude the subsequent use of specified materials in effecting repairs or replacements. Requests for approval of such substitutions shall be submitted to contracting officer and shall be accompanied by completely detailed supporting data which clearly illustrates the use of the proposed substitute material.

3.5 Interchangeability of components. Jigs, templates, gages, and fixtures shall be used to insure interchangeability of like components and subassemblies with respect to assembly, performance, and quality.

3.6 Adjustments. All parts and components subject to adjustment or replacement by operating or maintenance personnel shall be accessible. Adjusting facilities where provided shall operate without interference and shall maintain such adjustments under all conditions of service.

3.7 Tolerances and fits. Tolerances and fits shall conform to all appropriate AAR manuals applicable to railway cars unless otherwise specified herein.

3.8 Dimensions and weight. Dimensions and weight of cars shall be as follows:

| | |
|---|---|
| Length over striking plates | 58'-6" (nominal) |
| Length of flat depressed section | 18'-0" |
| Width of car | 9'-0" |
| Height from rail to top of bed of depressed center section (light car conditions) | 2'-0" (maximum) |
| Truck centers | 40'-0" (nominal) |
| Lightweight of car (figure 1) | 130,000 lbs. (maximum) |
| Lightweight of car (figure 2) | 132,500 lbs. (maximum) |



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3.9 Clearances. Car shall not exceed clearance limitations of AAR Plate B for unrestricted Interchange Service. Car with equipment (figure 3) shall be within clearance limitations of AAR Plate C.

3.9.1 Curvature. The car shall be capable of negotiating a 125-foot radius curve when uncoupled. Car shall be capable of negotiating a 125-foot radius curve when coupled by an auxiliary type coupler similar to National 31629 to a locomotive equipped with pilot type coupler (see 4.5.2.6).

3.10 Design and construction. The car shall conform to the AAR Manual of Standards and Recommended Practices and its supplement. Longitudinal and transverse members shall be of sufficient strength to support and transmit loads as specified in AAR Rules Governing the Loading of Commodities on Open Top Cars. Structural members and components shall have load carrying capacity to resist the design loads and superimposed dynamic loads as specified without exceeding the yield strength of the material. Structural members shall be free of notch sensitive areas and discontinuities that could result in high localized stress concentrations. The car shall be designed for the following load carrying capacity:

(a) 75 percent of the load concentrated on a knife edge at center of car.
(b) 100 percent of load uniformly distributed over 18 feet measured longitudinally, 9 feet on each side of transverse centerline of car.
(c) Under light car conditions clearance from the top of the rail to the underside of the lowest point of the depressed center structure shall not be less than 7-3/8 inches.
(d) The car shall carry the load when 140 tons is equally distributed over the mounting pads in car bed as indicated on figure 1 or figure 2, sheets 1 and 2.
(e) When specified, underframe shall be furnished with pad area of eight tapped holes (figure 2, sheet 1 and 2) for a vertical load of 300,000 pounds uniformly distributed over 10 feet at longitudinal centerline of car.

3.11 Underframe. The underframe shall be a one-piece steel casting conforming to AAR M-201, grade A, with an 18-foot depressed center section, center filler, draft lugs, body center plate, and strikers cast integral.

3.11.1 Pad areas. The pad areas in car bed shall be in accordance with one of the following figures as specified:

(a) Figure 1. Eight pads with four access holes.
(b) Figure 2. Two pads with eight tapped holes.



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3.12 Decking. No decking shall be applied to the depressed center section of the car. All other platform sections shall be covered with 1/4-inch raised diamond antiskid steel floor plate conforming to ASTM A36. The floor plate shall be welded to the underframe.

3.13 Couplers and yokes. Couplers shall be AAR type E, B-E68-HT with 31-inch "F" shank, bottom operated with coupler operating levers at diagonal corners. Yokes shall be of AAR design. One auxiliary coupler similar to National 31629 shall be supplied for test purposes and shall be shipped with the last car of the order.

3.14 Draft gear. Draft gear shall be AAR certified type for 24-5/8-inch pocket.

3.15 Routing card board. A standard AAR routing card board shall be applied to a side sill.

3.16 Defect card holder. A standard AAR metal defect card holder shall be applied to the car.

3.17 Safety appliances. Safety appliances shall conform to United States Safety Appliances for All Classes of Cars and Locomotives.

3.18 Brake system. The car shall be equipped with dual AB1012 brake systems with ABD control valves and automatic slack adjusters. Systems shall conform to the AAR Manual of Standards and Recommended Practices, Section E. Schematic brake diagram indicating lever arrangement and braking forces should be furnished to the contracting officer prior to fabrication of badge plate. Installation shall be in accordance with Specification No. 2518 as adopted by the AAR.

3.18.1 Slack adjusters. Automatic slack adjusters, for each brake system, shall be AAR approved and shall have a range of travel which will permit brake shoes to clear wheel tread surface with all shoes worn to limit or with all shoes new.

3.18.2 Handbrakes. Handbrakes shall be AAR approved, nonspin, drop shaft, quick release type. A 125-pound force applied to the handbrake shall develop a force on the brake shoe equal to the force of the airbrakes. Each car shall be equipped with two handbrakes; one applied at each side of the car in diagonal locations. Handbrakes shall not foul the underframe or other components of the car.

3.18.3 Badge plate. A badge plate for each brake system on the car conforming to the AAR Manual of Standards and Recommended Practices, Section E,



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shall be furnished. Badge plates for each applicable system shall be located over each truck on one of the side sills of the car.

3.19 Trucks. Trucks shall be AAR approved design for 7- by 12-inch journals with roller bearing and 56-1/2-inch gage track. The trucks shall have built-in snubbers and external hydraulic snubbing devices arranged to dampen vertical and lateral accelerations.

3.19.1 Springs. Springs shall be AAR D-4 design in accordance with section D of the AAR Manual of Standards and Recommended Practices.

3.19.2 Wheels. Wheels shall be BX33 multi-wear, wrought steel in accordance with Section G of the AAR Manual of Standards and Recommended Practices. Material shall conform to AAR M-107, class BR. Wheels shall be mounted in accordance with the AAR Wheel and Axle Manual.

3.19.3 Axles. Axles shall have 7- by 12-inch journals for roller bearings and raised wheel seats with journal and wheel fit dimensions in accordance with the AAR Wheel and Axle Manual. Material shall conform to AAR M-101. Ends of axles shall be drilled and tapped for roller bearing retaining caps.

3.19.4 Axle bearings. Axle bearings shall be roller type, AAR approved design. Bearing adapters shall be the narrow pedestal type conforming to the AAR Manual of Standards and Recommended Practices.

3.20 Side bearings. Side bearings shall be the integral cast type with adjustable shim plates.

3.21 Lubrication. Roller bearings shall be grease lubricated, using grease conforming to AAR M-917, and applied in quantities approved by the bearing manufacturer. Truck center plates shall be lubricated in accordance with AAR Recommended Practices.

3.22 Identification plates. An identification plate conforming to MIL-P-3558, type II, shall be permanently attached to the side sill of the car in a visible location.

3.23 Painting and marking. Unless otherwise specified, painting shall be in accordance with MIL-P-3320, table I, for Army domestic car requirements. Marking shall be as specified in MIL-P-3320 and as required by the AAR for cars in interchange service. Registration numbers shall be as specified.



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3.23.1 Automatic car identification (ACI) labels. ACI label shall be applied to each side of the car. Location shall be submitted to the contracting officer for approval.

3.24 Workmanship.

3.24.1 Components. All components shall be within manufacturing tolerances to permit proper fitting during fabrication. Components shall be free from discontinuities, notches, cracks, die marks, welding slag or undercuts, burrs and any other injurious defects. Shims, spacers, caulking compound, weld build-up shall not be used to make allowances for poor fitting components. Start and finish of hand and automatic welds shall be done in a manner to provide uniform welds. End weld metal in the high-stressed members shall be feathered into the parent metal of the parts being connected.

3.24.2 Welding. The surface of all parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Unless otherwise specified herein, welds shall develop the strength as required for the parts connected. All welding shall be in accordance with the latest applicable codes of the American Welding Society. Welders shall be qualified in accordance with the Qualification Test for welding operators as contained in section 6 of the AAR Electrical Manual of Standards and Recommended Practices.

3.24.3 Bolted and riveted connections. All bolt and rivet holes shall be accurately punched or drilled and shall have the burrs removed. Mating surfaces shall be flat and parallel and matching holes shall be concentric. Rivets shall be driven to completely fill the holes and each type head of the same diameter rivets shall be uniform in size and shape. All bolts, nuts, and screws shall be tight. Washers or lockwashers shall be provided in accordance with standard railway practice.

3.24.4 Assembled car. The assembled car shall be free from any defects that could cause malfunctioning of components or interfere with its safe operation. Construction of the car and appearance of the applied coatings shall be compatible with standard quality workmanship for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the



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specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Components and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with applicable AAR specifications, or that the components have been previously approved by AAR.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First-produced car(s) inspection (see 4.3).
(b) Quality-conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 First-produced car(s) inspection. The first-produced car(s) which shall have been produced from production tooling shall be examined and tested to determine conformance to the requirements of this specification.

4.3.1 Examination. The first-produced car(s) shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Tests. The first-produced car(s) shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each car shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each car shall be tested as specified in 4.5.2.1 through 4.5.2.5. Failure of any test shall be cause for rejection.

4.5 Inspection procedures.

4.5.1 Examination. All cars shall be examined for compliance with the specification. This examination shall include the following:

101. Material not as specified herein.
102. Design and construction not as specified herein.
103. Dimensions, tolerances, and fits not as specified herein.
104. Parts and components missing, misalined, or not as specified.
105. Brake systems and components not as specified.
106. Pipe fittings misalined, missing or not tight.



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107. Adjustments not as specified.
108. Threaded parts not as specified.
109. Safety appliances not as specified herein.
110. Painting and marking not as specified.
111. Lubrication not as specified.
112. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to being tested, each car shall be lubricated with oil and grease as specified herein and put in a completely operational condition. Methods of performing tests shall be submitted to the contracting officer for approval.

4.5.2.2 Airbrake. The car shall be tested with a single car testing device in accordance with Instruction Pamphlet No. 5039-4, Supplement 1, adopted by AAR. Nonconformance to 3.18 or failure of this test shall be cause for rejection.

4.5.2.3 Handbrake. The handbrake shall be tested by applying and releasing the brakes to determine that brake shoes are applied to all wheels and that the operation conforms to 3.18.2. Nonconformance to 3.18.2 shall constitute failure of this test.

4.5.2.4 Side bearing clearance. With the car on level tangent track and the car body level, side bearing clearance shall be measured at each bearing. Nonconformance to AAR standard limits shall constitute failure of this test.

4.5.2.5 Slack adjuster. The slack adjuster shall be tested for range of travel. Nonconformance to 3.18.1 shall constitute failure of this test.

4.5.2.6 Curvature. The car, uncoupled, shall be moved through a 125-foot radius curve. At maximum swing, the brake rigging and the car body rigging shall be inspected to determine that clearances have been provided and the brake systems function as specified. A car coupled to a locomotive by means of an auxiliary coupler similar to National 31629 shall be moved through a 125-foot radius curve to determine that the coupled units can negotiate the curve. Nonconformance to this test or 3.9.1 shall constitute failure.

4.5.2.7 Load. The car shall be loaded in accordance with design loads specified in 3.1 and 3.10. There shall be no permanent resulting negative deflection in frame or unbalance of deck. Nonconformance to 3.1 or 3.10 shall constitute failure of this test.



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4.5.2.8 Load for cast steel underframe. The underframe shall be tested by loading in a manner to produce a bending moment of two times as great as the maximum bending moment produced by the design loads specified in 3.1 and 3.10. Nonconformance to AAR requirements of design and construction shall constitute failure of this test.

4.5.2.9 Tapped holes. When pads with eight tapped holes are furnished, a force of 90,000-100,000 pounds to each of the eight 2.75-8 tapped holes shall be applied for a minimum of 10 minutes. Evidence of cracks or deformation after loading shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Examination. Each car shall be examined for the following defects:

113. Car not prepared for delivery as specified.

5. PREPARATION FOR DELIVERY

5.1 The cars shall be prepared for delivery in accordance with AAR requirements for interchange service.

6. NOTES

6.1 Intended use. The domestic service depressed center flatcar is intended to transport a shipping container which will be attached to the flatcar by the Government.

6.2 Ordering data. Procurement documents should specify the following:

(c) Title, number, and date of this specification.
(b) Number of first-produced cars to be furnished and time frame required for submission of first-produced cars (see 3.2.1).
(c) When underframe shall be designed for a load of 300,000 pounds (see 3.10).
(d) Pad area in car bed to be required (see 3.11.1).
(e) When painting is to be other than specified (see 3.23).
(f) Registration numbers to be furnished (see 3.23).

6.3 First-produced car. Any changes or deviations from the approved first-produced car(s) during production will be subject to approval of the contracting officer. Approval of a first-produced car(s) by the Government shall not relieve the supplier from his obligation to furnish cars conforming to this specification.

Custodian:

Army - ME

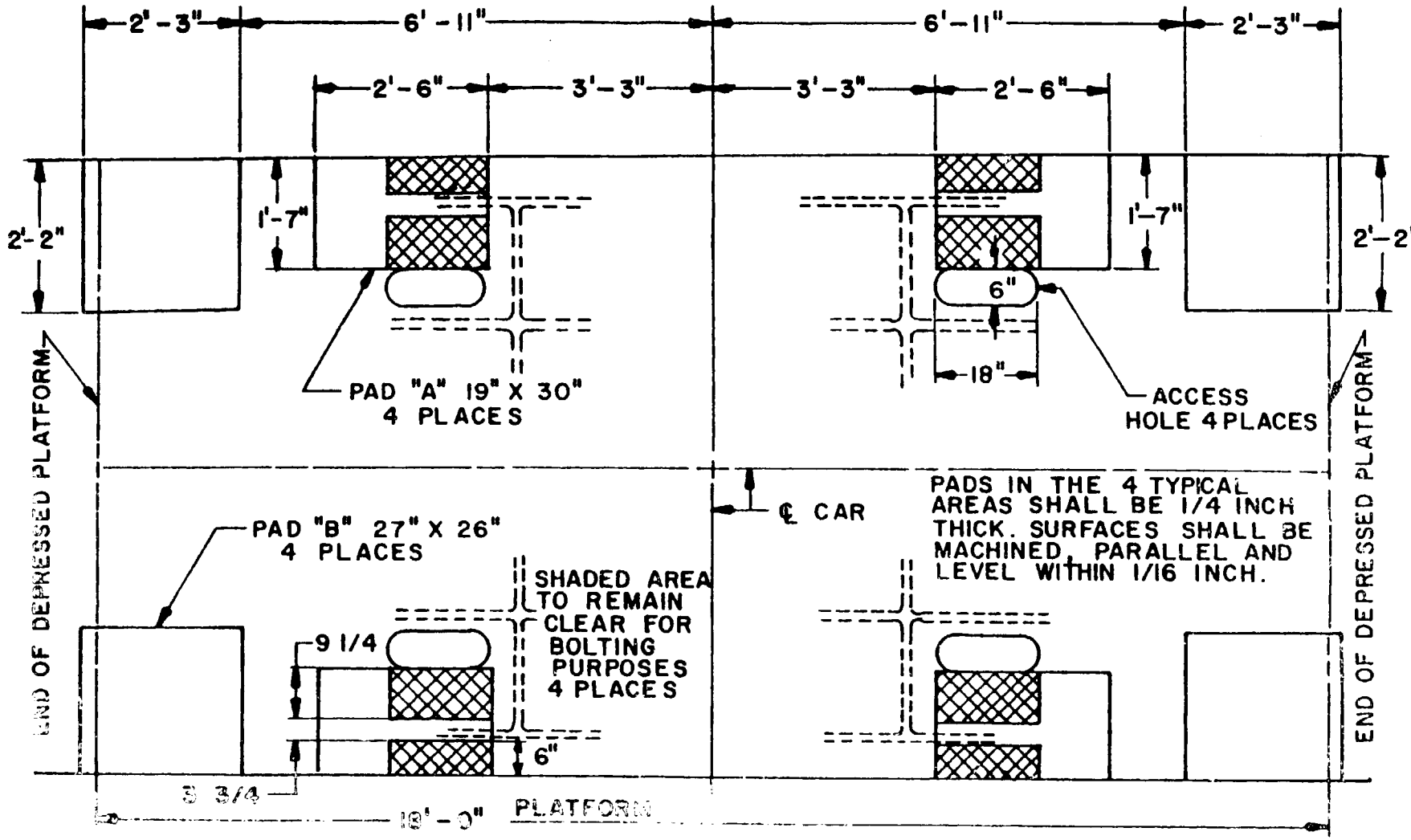
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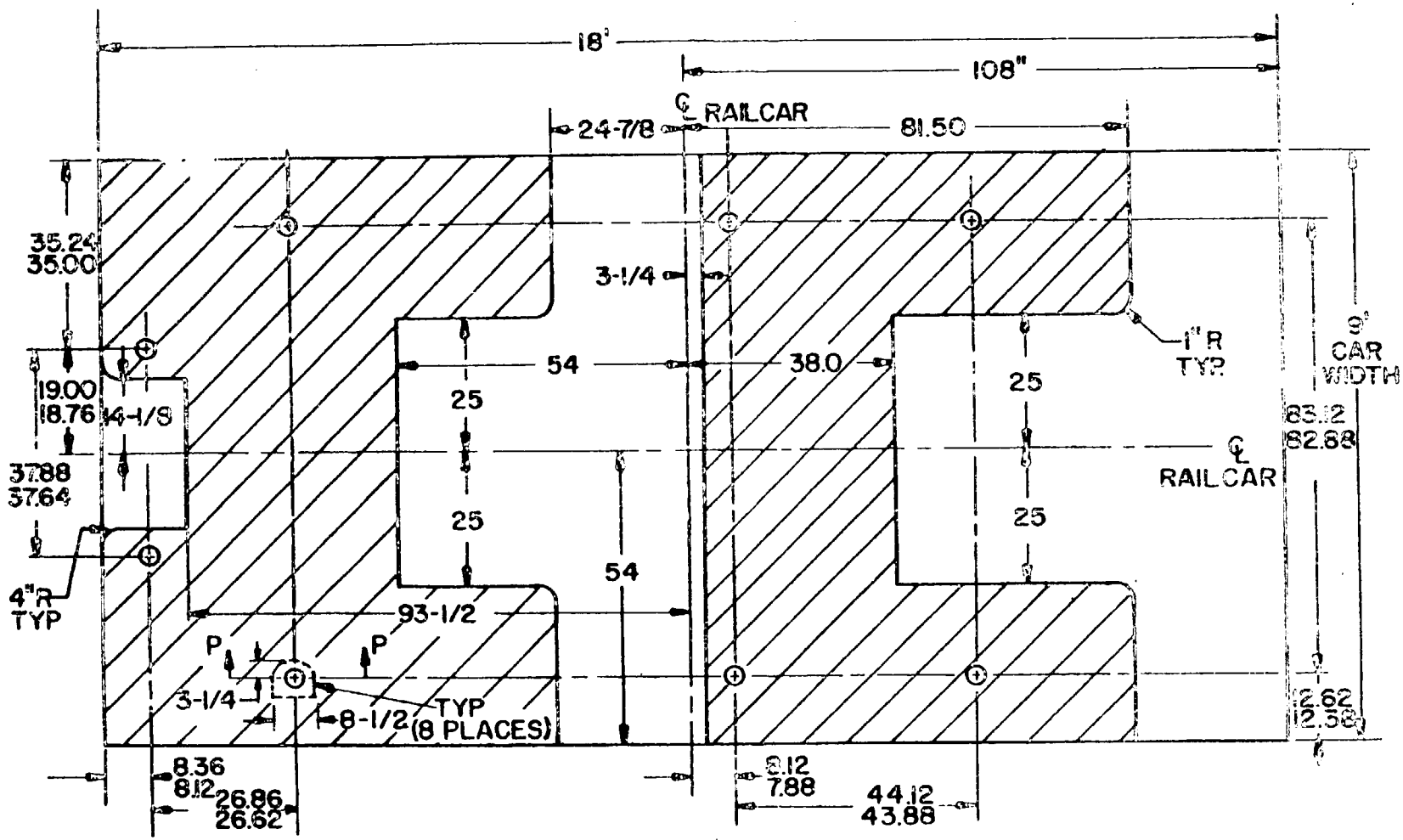

LOCATION OF PADS AND ACCESS HOLES ON PLATFORM OF
140 TON DEPRESSED CENTER FLAT CAR

FIGURE 1

X-2202



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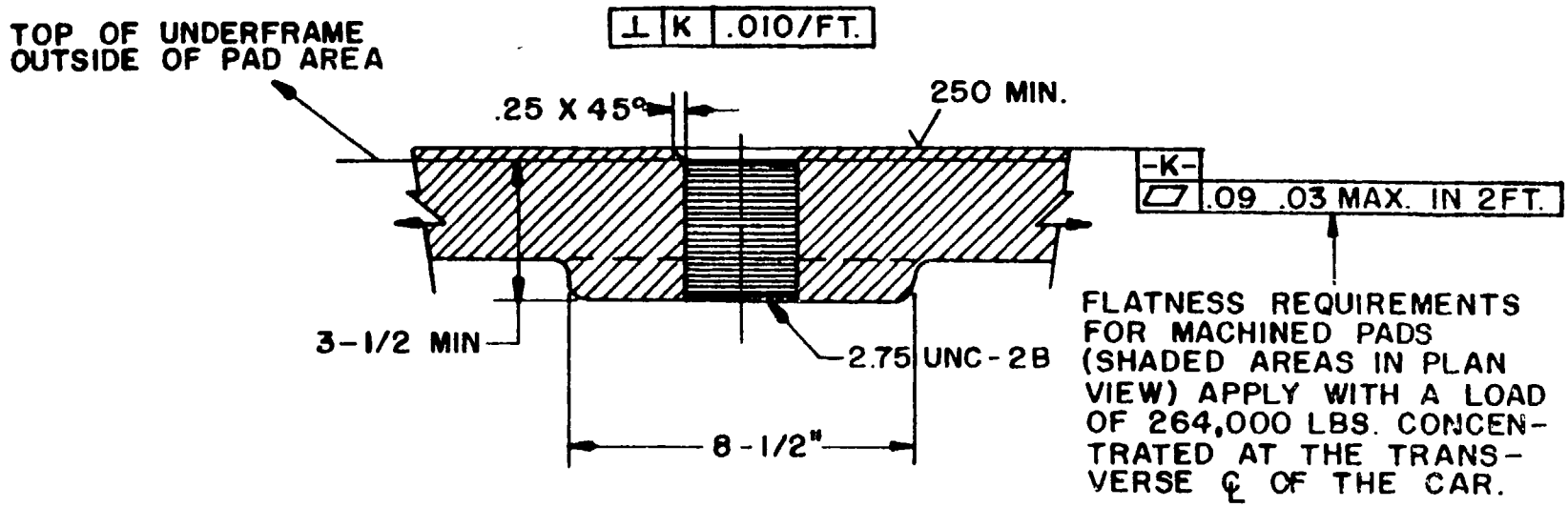

LOCATION OF PADS (WITH 8 TAPPED HOLES) ON PLATFORM,
140 TON DEPRESSED CENTER FLAT CAR
FIGURE 2

SHEET 1 OF 2

X-2203



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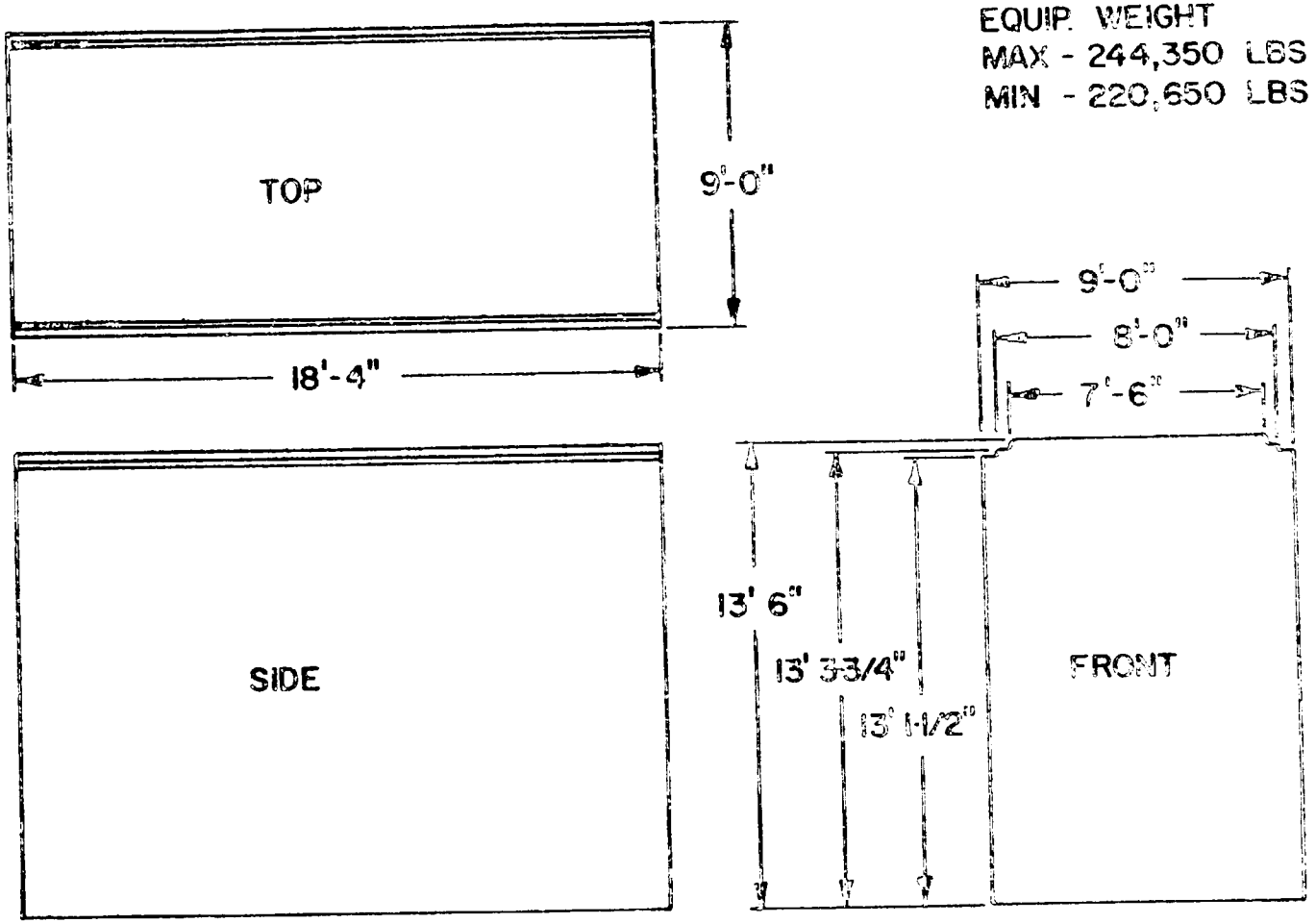



ENLARGED SECTION P-P
FIGURE 2

(14)



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EQUIPMENT SPACE ENVELOPE

FIGURE 3

X-2205

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*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. [illegible] TELEPHONE NUMBER [illegible] |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | d. [illegible] |

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